Experiment – 7: Continuous-Time Fourier Transform and Spectrum.

OBJECTIVES:

(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

(iii) To verify important Fourier transform properties and examine their effects on the corresponding time-domain and frequency-domain representations.

(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc;
clear;
close all;

% Experiment 7(ii): Magnitude and Phase Spectra

t = -2:0.001:2;
dt = t(2)-t(1);

x = exp(-t.^2).*cos(2*pi*3*t);

N = 4096;
X = fftshift(fft(x,N))*dt;
```

```
f = (-N/2:N/2-1)/(N*dt);

magX = abs(X);
phaseX = unwrap(angle(X));

% Time-domain signal
figure;

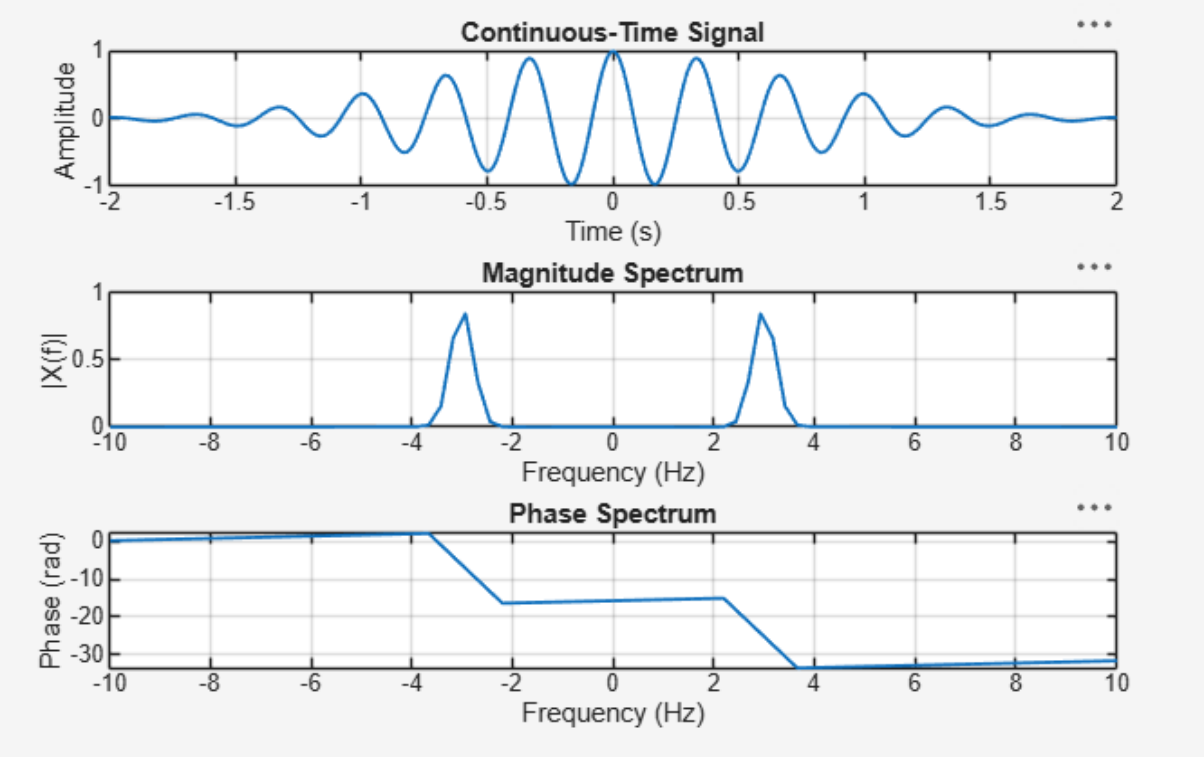
subplot(3,1,1);
plot(t,x,'LineWidth',1.1);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Continuous-Time Signal');

% Magnitude spectrum
subplot(3,1,2);
plot(f,magX,'LineWidth',1.1);
xlim([-10 10]);
grid on;
xlabel('Frequency (Hz)');
ylabel('|X(f)|');
title('Magnitude Spectrum');

% Phase spectrum
subplot(3,1,3);
```

```
plot(f,phaseX,'LineWidth',1.1);
xlim([-10 10]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Phase Spectrum');
```

OUTPUT:



(iii) Verification of Fourier Transform Properties and Frequency-Domain Characteristics.

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc;
```

```
clear;
close all;

% Experiment 7(iii): Verification of Fourier Transform Properties

t = -2:0.001:2;
dt = t(2)-t(1);
t0 = 0.4;

x = exp(-t.^2);

% Time-shifted signal
x_shift = interp1(t,x,t-t0,'linear',0);

% FFT with zero-padding
N = 8192;
X = fftshift(fft(x,N))*dt;
X_shift = fftshift(fft(x_shift,N))*dt;

% Frequency axis
f = (-N/2:N/2-1)/(N*dt);

% Theoretical time-shift property
X_shift_theory = X.*exp(-1j*2*pi*f*t0);

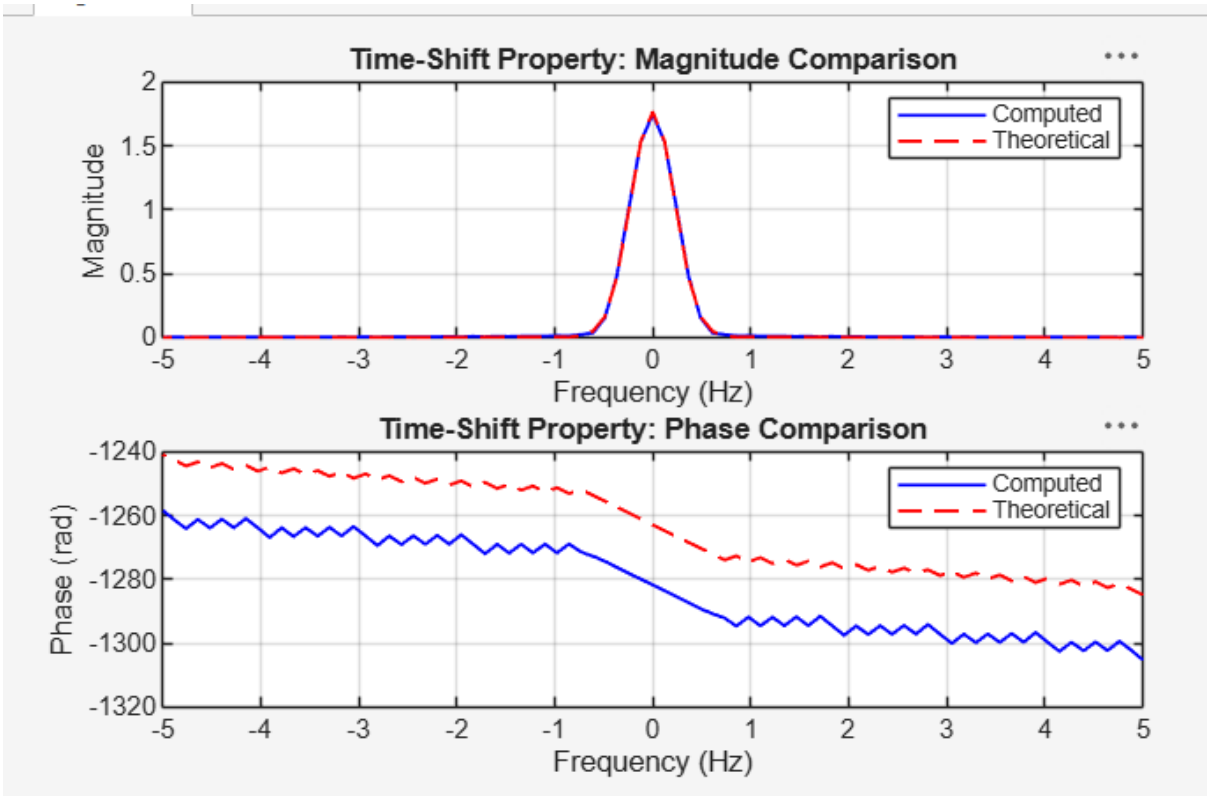
% Magnitude comparison
```

```
figure;

subplot(2,1,1);
plot(f,abs(X_shift),'b',f,abs(X_shift_theory),'r--','LineWidth',1.1);
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');

% Phase comparison
subplot(2,1,2);
plot(f,unwrap(angle(X_shift)),'b', ...
    f,unwrap(angle(X_shift_theory)),'r--','LineWidth',1.1);
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Time-Shift Property: Phase Comparison');
legend('Computed','Theoretical');
```

Output Graphs:



RESULT:

The continuous-time Fourier transform of representative signals was successfully obtained using FFT-based approximation, and their magnitude and phase spectra were analyzed to identify significant frequency components. The time-shift property was verified by comparing the computed and theoretical Fourier transforms, confirming that time shifting preserves magnitude while introducing a frequency-dependent phase shift. The spectral components and occupied bandwidth of a composite communication signal were also determined, demonstrating the importance of frequency-domain analysis in communication-system design.

RUBRICS:

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks(10)**	**Marks obtained**

(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.	**Fourier Transform Computation & MATLAB Implementation**	Computes X(jω) of every given CT signal correctly, analytically and numerically . Uses a suitable time range, sampling rate and frequency axis in rad/s or Hz.	Computes the spectra correctly, with minor errors in scaling, sampling rate, frequency axis or parameters.	Computes the spectra for only some signals, or has noticeable errors in code, scaling or frequency axis, and needs assistance.	Unable to compute the Fourier transform or produce meaningful spectra.	**20**	2	
	Magnitude & Phase Spectrum Plots	Plots |X(jω)| and ∠X(jω) correctly for every signal, with	Plots both spectra correctly, with minor errors in scaling,	Plots the spectra with noticeable errors in scaling	Unable to plot the magnitude and phase spectra.	**20**	2	

		proper scaling, titles, axis labels and grid. Handles phase wrapping and phase near zero-magnitude points correctly.	labelling, phase wrapping or axis range.	, phase handling or labelling, and needs guidance.				
	Spectral Component Analysis	Correctly identifies dominant frequencies, peaks, bandwidth and the frequency range holding most	Identifies most significant spectral components and bandwidth, with minor errors or limited justification.	Identifies only basic spectral components. Bandwidth or symmetry reasoning is incomplete, and needs	Unable to identify or justify the significant spectral components.	**20**	2	

		of the signal energy. Uses symmetry (even magnitude, odd phase for real signals) and relates the spectral lines or lobes to the signal parameters.		guidance.				
	Analytical vs Numerical Verification	Compares the analytical and MATLAB spectra, quantifies the difference	Compares analytical and numerical results correctly, with brief explanation of	Makes a basic comparison with noticeable errors, and the explanation of	Unable to verify the numerical spectra against the analytical results.	**20**	2	

		and explains its causes such as truncation, sampling and spectral leakage. Shows how signal parameters change the spectrum.	differences.	differences is incomplete.				
	Engineering Analysis, Record & Interpretation	Provides technically accurate interpretation of the results and relates them to the	Maintains a proper record with correct interpretation, with minor omissions in explanation or	Record and interpretation are partial or only descriptive, and need guidance.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	

		physical or engineering meaning of the signal. Maintains a complete, well-organized record with observations, tables, plots and conclusions.	conclusions.					
					Total	**100**	**10**	